Yixin (Stephanie) Zha

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Education

Bachelor of Applied Science in Electrical & Computer Engineering Toronto, ON

University of Toronto | Expected May 2027
GPA: 3.94 / 4.0
Relevant Coursework: Deep Neural Networks with PyTorch, Applied Machine Learning, Algorithms & Data Structures, Computer Organization, Digital Systems, Software Design

Skills

Languages: C++, Python, SystemVerilog, Verilog, JavaScript, Assembly, SQL, HTML/CSS
ML / Systems: PyTorch, MLIR, MNN, TensorFlow Lite, Keras, SnnTorch, NumPy, Pandas
Hardware / Tools: DE1-SoC FPGA, Intel Quartus, ModelSim, Git, VS Code, SQLite, LTspice

Professional Experience

Machine Learning Compiler Engineer - Cerebras Systems May 2025 - Aug 2026

- Developed PyTorch frontend compiler pipelines for Cerebras wafer-scale hardware using MLIR, including dialects, rewrite patterns, and shape/type inference for hardware-aware lowering.
- Contributed to quantization, prompt caching, model tracing, and other engine features.
- Supported GPT, LLaMA, and Qwen model bring-up and validation; debugged correctness, performance, and numerical issues across PyTorch, MLIR, and hardware execution.
- Contributed skills and cross-disciplinary knowledge of simulation testing, wafer architecture, and engine parameters to support agentic workflows for model bring-up and debugging.

Machine Learning Engineer Intern - Shanghai Legend Energy Ltd. Apr 2023 - Aug 2023

- Performed time-series preprocessing and feature analysis with Pandas, improving model accuracy by 20%.
- Evaluated and deployed 10+ ML models (LR, RF, XGBoost, VAE), improving prediction accuracy from 94% to 99.8%.
- Built unsupervised anomaly detection models achieving 80% fault detection accuracy.

Research Experience

Research Team Lead - TinyViT FPGA Implementation Feb 2026 - Present

University of Toronto - Professor Vaughn Betz

- Developing an end-to-end TinyViT inference pipeline on the DE1-SoC FPGA board.
- Designing a systolic array to accelerate multiply-accumulate (MAC) operations.
- Implementing a custom MNN backend to integrate custom SystemVerilog modules into the inference pipeline.

Research Assistant - Edge ML for Sleep Disorder Therapy May 2025 - Aug 2026

University of Toronto - Professor Xilin Liu

- Designed edge ML models for sleep disorder treatment and closed-loop neuromodulation; optimized TensorFlow Lite models for low-latency, low-power inference on Nordic nRF microcontroller boards.

Research Assistant - Spiking Neural Networks Apr 2024 - Aug 2024

University of Toronto - Professor Roman Genov

- Researched spike-timing-dependent plasticity (STDP); implemented SNNs using PyTorch and SnnTorch, achieving 99.2% accuracy on MNIST.

Projects

University Mapping GIS Software (C++): Designed a commuter-oriented GIS application using OpenStreetMap; led team development and Git workflow.
Real-Time Object Detection (YOLOv8): Implemented YOLOv8 models in Python for real-time detection.
FPGA Pong Game (Verilog): Built a two-player VGA Pong game using FSM-based control and datapath.

Awards

Dean's List (2022-2025) | University of Toronto Scholar Award (2023)
ECE Outstanding Student Award (Top 2) | Intel Andy Grove Scholarship (2023)